GSE 518
Prof. Carlos A. Flores
Cal Poly State University

Suggested Solutions to Problem Set 4

1. Remember that by definition $E(X) = \int xf(x)\,dx$, and $Var(X) = \int (x - E(x))^2 f(x)\,dx$ for continuous X. For discrete X we use summation instead of integration. Remember that for the variance is usually easier to calculate $Var(X) = E(X^2) - [E(X)]^2$.

(a) $E(X) = \int_0^1 xax^{a-1}dx = \int_0^1 ax^a dx = \frac{ax^{a+1}}{a+1}\Big|_0^1 = \frac{a}{a+1}$

$E(X^2) = \int_0^1 x^2 ax^{a-1}dx = \int_0^1 ax^{a+1}dx = \frac{ax^{a+2}}{a+2}\Big|_0^1 = \frac{a}{a+2}$

$Var(X) = \frac{a}{a+2} - \left[\frac{a}{a+1}\right]^2 = \frac{a}{(a+2)(a+1)^2}$

(b) For this problem we need to use the results: $\sum_{i=1}^k i = \frac{k(k+1)}{2}$ and $\sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$.

$E(X) = \sum_{x=0}^n \frac{x}{n} = \frac{1}{n}\sum_{x=0}^n x = \frac{1}{n}\frac{n(n+1)}{2} = \frac{n+1}{2}$

$E(X^2) = \sum_{x=0}^n \frac{x^2}{n} = \frac{1}{n}\sum_{x=0}^n x^2 = \frac{1}{n}\frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}$

$Var(X) = \frac{(n+1)(2n+1)}{6} - \left[\frac{n+1}{2}\right]^2 = \frac{2n^2+3n+1}{6} - \frac{n^2+2n+1}{4} = \frac{n^2-1}{12}$

2.

(a) $\int_{-\infty}^{\infty} f(x)\,dx = \int_1^{\infty} \frac{2}{x^3}dx = -\frac{1}{x^2}\Big|_1^{\infty} = 0 - (-1) = 1$

(b) $E(X) = \int_1^{\infty} x\frac{2}{x^3}dx = \int_1^{\infty} \frac{2}{x^2}dx = -\frac{2}{x}\Big|_1^{\infty} = 0 - (-2) = 2$

(c) $E(X^2) = \int_1^{\infty} x^2\frac{2}{x^3}dx = \int_1^{\infty} \frac{2}{x}dx = 2\ln(x)|_1^{\infty} = \infty$. Thus, $Var(X)$ does not exist.

3.

(a) The approach here is very similar to the one we followed in class to show that we can write the variance of a random variable X as $Var(X) = E(X^2) - [E(X)]^2$. Expanding the cubic inside the expectation we find $E[(X-\mu)^3] = E[X^3 - \mu^3 + 3X\mu^2 - 3X^2\mu] = E(X^3) - \mu^3 + 3\mu^2 E(X) - 3\mu E(X^2) = E(X^3) - 3\mu E(X^2) + 2\mu^3$.

(b) In this case, we have that:

$E(X) = \int_{-1}^1 x\frac{1}{2}dx = \frac{x^2}{4}\Big|_{-1}^1 = \frac{1}{2} - \frac{1}{2} = 0$, so $\mu = E(X) = 0$. And,

$E(X^3) = \int_{-1}^1 x^3\frac{1}{2}dx = \frac{x^4}{8}\Big|_{-1}^1 = \frac{1}{8} - \frac{1}{8} = 0$

Thus, $E[(X-\mu)^3] = E(X^3) = 0$ and $E[(X-\mu)^3]/\sigma^3 = 0$. This should not be a surprise since this $f(x)$ is the pdf of a uniform distribution, $U(-1,1)$, which we know is symmetric.

(c) Note that since for the distribution in part (b) we have $E(X) = 0$, then the measure of kurtosis reduces to $E[(X-\mu)^4]/\sigma^4 = E(X^4)/[E(X^2)]^2$. In this case we have

$E(X^2) = \int_{-1}^1 x^2\frac{1}{2}dx = \frac{x^3}{6}\Big|_{-1}^1 = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

$E(X^4) = \int_{-1}^1 x^4\frac{1}{2}dx = \frac{x^5}{10}\Big|_{-1}^1 = \frac{1}{10} + \frac{1}{10} = \frac{1}{5}$

Therefore, $E[(X-\mu)^4]/\sigma^4 = \left(\frac{1}{5}\right)\left(\frac{1}{3}\right)^{-2} = \frac{9}{5}$

4. The mgf is given by

$M_X(t) = E\left(e^{tX}\right) = \int_a^b e^{tx}\left(\frac{1}{b-a}\right)dx = \frac{1}{t(b-a)}e^{tx}\Big|_a^b = \frac{1}{t(b-a)}\left(e^{tb} - e^{ta}\right)$

Note that in this case the mgf is not defined at $t = 0$, and therefore, it cannot be used directly to derive the moments of a uniformly distributed random variable.

The mean and variance are given by

$E(X) = \int_a^b x\left(\frac{1}{b-a}\right)dx = \frac{x^2}{2(b-a)}\Big|_a^b = \frac{b^2}{2(b-a)} - \frac{a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \frac{b+a}{2}$

$E(X^2) = \int_a^b x^2\left(\frac{1}{b-a}\right)dx = \frac{x^3}{3(b-a)}\Big|_a^b = \frac{b^3}{3(b-a)} - \frac{a^3}{3(b-a)} = \frac{(b-a)\left(b^2+ab+a^2\right)}{3(b-a)} = \frac{b^2+ab+a^2}{3}$

$Var(X) = \frac{b^2+ab+a^2}{3} - \left(\frac{b+a}{2}\right)^2 = \frac{(b-a)^2}{12}$

5.

(a) Using the definition of the moment generating function and the pdf of a normal we have

$$\begin{aligned} M_X(t) &= E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/2\sigma^2}dx \\ &= \int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}+tx}dx \\ &= \int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}\left[(x-\mu)^2-2\sigma^2tx\right]}dx \qquad (1) \end{aligned}$$

The easiest way to deal with this integral is to get something inside which is the pdf of a normal random variable, so that it integrates to one. To do so, we work with the term $(x-\mu)^2 - 2\sigma^2tx$ in the exponent of e to get something that looks like $(x-\delta)^2$, for some function δ of t. Then, whatever is left inside the integral is the pdf of a normal mean δ and variance σ^2. Continuing working with equation (1) we find

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}\left[(x-\mu)^2-2\sigma^2tx\right]}dx \\ &= \int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}\left[(x-\mu)^2-2\sigma^2tx+2\sigma^2t\mu-2\sigma^2t\mu\right]}dx \\ &= \int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}\left[(x-\mu)^2-2\sigma^2t(x-\mu)-2\sigma^2t\mu\right]}dx \\ &= \int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}\left[(x-\mu)^2-2\sigma^2t(x-\mu)\right]}e^{t\mu}dx \\ &= e^{t\mu}\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}\left[(x-\mu)^2-2\sigma^2t(x-\mu)+\left(\sigma^2t\right)^2-\left(\sigma^2t\right)^2\right]}dx \\ &= e^{t\mu}\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}\left[(x-\mu)^2-2\sigma^2t(x-\mu)+\left(\sigma^2t\right)^2\right]}e^{\frac{t^2\sigma^2}{2}}dx \\ &= e^{t\mu+\frac{t^2\sigma^2}{2}}\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2\sigma^2}\left[\left(x-\mu-\sigma^2t\right)^2\right]}dx \\ &= e^{t\mu+\frac{t^2\sigma^2}{2}} \end{aligned}$$

Since the function inside the integral is the pdf of a normal distribution mean $\mu + \sigma^2 t$ and variance σ^2, we know it integrates to 1. Therefore, $M_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}}$.

(b) The first two derivatives of $M_X(t)$ with respect to t are given by $M'_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}}(\mu + t\sigma^2)$ and $M''_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}}(\mu + t\sigma^2)^2 + e^{t\mu + \frac{t^2\sigma^2}{2}}(\sigma^2)$. Evaluating these two derivatives at zero we have (see Theorem 2.3.7 in CB): $E(X) = M'_X(0) = \mu$ and $E(X^2) = M''_X(0) = (\mu)^2 + \sigma^2$. Therefore, we have: $E(X) = \mu$, and $Var(X) = E(X^2) - [E(X)]^2 = \mu^2 + \sigma^2 - \mu^2 = \sigma^2$.